

Inversely Designed Ultrasmall Scrambling-Type Mode Multiplexer Based on Mosaic Structure

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Abstract We propose and experimentally demonstrate first scrambling-type mode multiplexers based on mosaic structure. Two- and three-mode multiplexers with an ultrasmall footprint of $< 10 \mu\text{m}^2$ are successfully demonstrated with the help of inverse design method based on gradient-based and machine learning techniques. ©2026 The Author(s)

Introduction

The extreme increase of data transmission capacity with AI requires significant capacity increase in optical fibre transmission system. A mode-division multiplexing (MDM) [1-3] enables a substantial increase in transmission capacity per fibre core and is considered a key technology for next-generation large-capacity optical communications. In MDM transmission, if the signals with different modes are mixed during the transmission, a MIMO technique is necessary at the receiver side. In that case, so-called, differential mode delay (DMD) and mode-dependent loss (MDL) make it difficult to recover the signal. To avoid these problems, mode exchanging technique was proposed [4-7]. For N_m -mode transmission, by placing N_m-1 mode exchangers (mode-EX) at the relay point of a few-mode fibre (FMF), at least theoretically, complete cyclic mode conversion can be achieved. Especially, integrable mode exchangers are useful due to their small size, and recently, ultrasmall size mode-EX was demonstrated using mosaic structure [7, 8]. Although this technique is useful for averaging MDL and DMD, multiple mode-EXs are necessary for complete mode exchanging.

In [9, 10], scrambling-type mode multiplexers (SC-mode MUXs) in silica waveguide platform was proposed. By using the SC-mode MUXs, the signals with different modes are mixed at the transmitter side, and DMD and MDL are averaged. Therefore, SC-mode MUX can help the mitigation of DMD and MDL in the MIMO-MDM link. However, due to the low-index-contrast of silica waveguide, the size of the devices was large ($> \text{mm}$), and they are not suitable for compact optical transmitter.

In this paper, we propose and experimentally demonstrate the smallest SC-mode MUXs in Si-photonics platform, for the first time. The use of mosaic structures makes it possible to control light with great flexibility as well as achieving

ultrasmall size. The complex mosaic patterns are obtained with an inverse design composed of gradient-based direct binary search (gDBS) combined with a machine learning technique based on Bayesian inference [11, 12]. Two- and three-mode SC-mode MUXs are successfully demonstrated with the ultrasmall size less than $10 \mu\text{m}^2$. Experimental results are in good agreement with the theory, showing the usefulness of the device.

Device structure

Figure 1 shows the device structures and working principles for two- and three-mode SC MUXs. We consider standard Si waveguide with the thickness of 220 nm with SiO₂ cladding. For two-mode case, it consists of two inputs and one output waveguides. A mosaic region is placed between them. The waveguide widths are set to $W_{\text{in}} = 0.6 \mu\text{m}$, $W_{\text{out}} = 0.8 \mu\text{m}$, with $L_z = 3.0 \mu\text{m}$, $L_x = 2.4 \mu\text{m}$, $W_{\text{gap}} = 1.0 \mu\text{m}$. The mosaic region is divided into 100 nm square pixels, and therefore, there are 720 pixels. Each pixel is designated as core or cladding. The device is intended to work, as shown in the top right panel of Fig. 1. The input is the TE₀ mode of the input waveguides, and at the output, TE₀ and TE₁ modes of the output waveguide are equally excited. Due to the operation symmetry, x-symmetry is imposed to the design of the mosaic patterns, resulting in 360 pixels in total.

The bottom panel of Fig. 1 shows the schematic and operation of three-mode case. For the three-mode, $W_{\text{in}} = 0.6 \mu\text{m}$, $W_{\text{out}} = 1.2 \mu\text{m}$, with $L_z = L_x = 3.0 \mu\text{m}$ (total 900 pixels), $W_{\text{gap}} = 0.5 \mu\text{m}$. In the ideal case, TE₀, 1, and 2 modes of the output waveguide is equally excited from each input. It should be noted that the SC mode MUX considered here is NOT the simple symmetric multi-mode power splitter [13, 14] for $N_m > 2$. For three-mode case, it is not possible to excite TE₁ mode from the center input (Port 2), and therefore,

nonsymmetric mosaic pattern is required, which is more difficult to design.

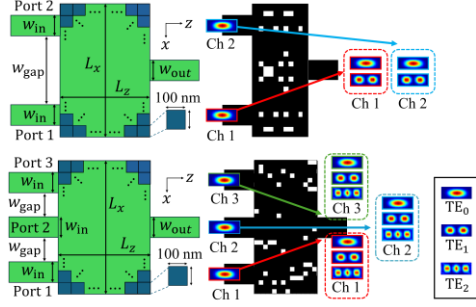


Fig. 1: Schematics of SC-mode MUXs and operation principles for (top) two-, and (bottom) three-mode case.

Inverse design method

In mosaic devices, how to find the appropriate mosaic pattern is the most important. Conventionally, the design has been performed using the direct binary search (DBS) method [15], in which a pixel is randomly selected, and its state is flipped. If the characteristic is improved through electromagnetic (EM) simulation, the inversion is retained, and if not, the pixel is returned to the original state. However, this method suffers from slow convergence and the tendency to be trapped in local optima. Moreover, as the number of pixels increases, the number of EM simulations is increased quadratically.

To address these issues, we developed gDBS method [8]. In this approach, the gradient in terms of target characteristics (figure of merit, FOM) with respect to each pixel is evaluated using the adjoint method [16, 17], and pixel states are updated based on this information. Compared to conventional DBS, gDBS enables faster convergence and provides a more physically grounded optimization process. This method is useful especially at the initial stage of the optimization. However, at the final stage of the optimization, conventional DBS was used to sweep all pixels [8]. During the gDBS optimization, the results of EM simulation are accumulated, and they can be used for the training data of machine learning technique. Therefore, we used Bayesian DBS (BDBS), which is also developed by our group, at the latter part of the optimization. Schematic image of the convergence compared with conventional DBS method is depicted in Fig. 2. In the first part of the optimization (where no training data exist), the gDBS is used to accelerate the convergence (optimization based on “physical” information of the system). At the same time, the training data are collected. In the latter part of the optimization, BDBS is used. In this stage, “statistical” optimization is done based on accumulated training data to accelerate the convergence further. This is the first demonstration of the combined use of gDBS and BDBS. For the EM simulation, we used finite-element method [18].

FOM is defined as

$$FOM = 1 - \frac{1}{N_m N_\lambda} \sum_i \sum_j \sum_k \left| T_{j,k}(\lambda_i) - \frac{1}{N_m} \right| \quad (1)$$

where N_λ is the number of wavelengths, N_m is the number of modes, $T_{j,k}$ is the transmission from Port j to TE_k mode at the output. The wavelengths for the optimization are set to 1530, 1550 and 1570 nm ($N_\lambda = 3$).

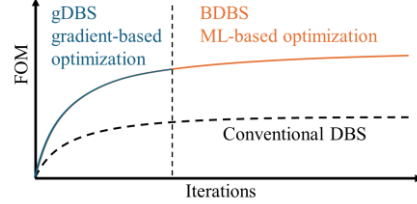


Fig 2: The schematic image of the FOM convergence as a function of iteration for gDBS+BDBS (solid) and conventional DBS (dashed).

Scrambling-type two-mode MUX

Here, the results of SC two-mode MUX are presented. Figure 3 shows the FOM as a function of iteration. For gDBS, multiple pixels can be simultaneously treated [8]. We perform two loops of gDBS iterations. 10 and 5 pixels are treated in the first and second gDBS iterations (therefore, 36 and 72 iterations are done). After that, three loops of BDBS optimisation are performed (1080 iterations). FOM is rapidly increased at gDBS stage and converges in the BDBS stage. The final pattern is shown in the inset of Fig. 3. Left and middle panels of Fig. 4 show the field distributions of the device for Port 1 and 2 inputs. The output is the mixture of TE_0 and TE_1 modes of the output waveguide.

The designed device is fabricated. An EB lithography is used for the patterning, and a dry etching technique is used, followed by the evaporation of the overcladding. The right panel of Fig. 4 shows the SEM picture of the fabricated device, showing the successful fabrication of complex mosaic patterns. We measured the device as DMUX, as shown in the inset of Fig. 5 (a micrograph of the device). Tunable lasers are used for the light source, and the light is coupled to the chip through a lensed fibre and spot size converter. The output light is measured by a power meter. For launching TE_1 mode, an asymmetric directional coupler (ADC) [19, 20] is placed before the device. TE_0 mode launched from the left side of lower input waveguide is converted to TE_1 mode at the ADC and inputted to the mosaic region. Solid and dashed lines in Fig. 5 show measured and calculated transmission spectra of the device. Top and bottom panels are for TE_0 and TE_1 inputs. For both inputs, the light is equally split to Port 1 and 2. Furthermore, measured and

calculated results are in excellent agreement, showing the validity of our designs.

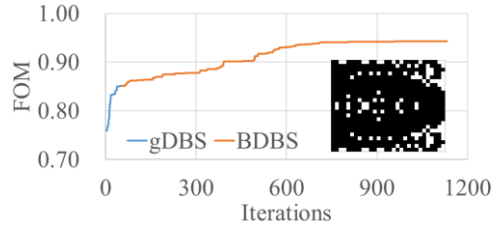


Fig 3: FOM as a function of SC two-mode MUX. Inset is designed mosaic pattern.

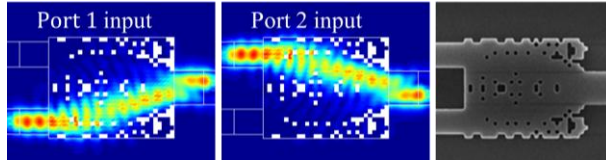


Fig 4: (Left, middle) Calculated field distributions for each input port and (right) SEM image of the fabricated device.

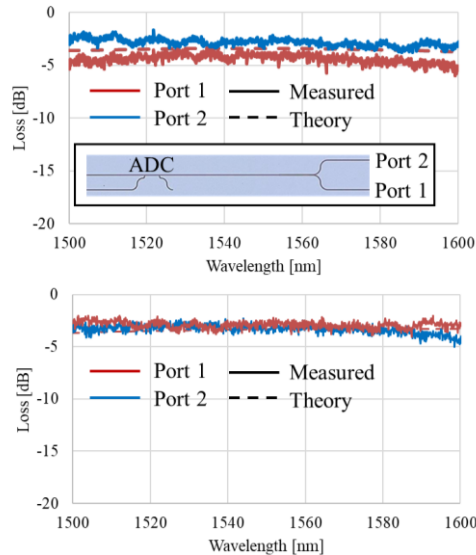


Fig 5: Measured and calculated transmission spectra of SC two-mode MUX for (top) TE_0 and (bottom) TE_1 input.

Scrambling-type three-mode MUX

Next, three-mode MUX is considered. As in the two-mode case, two loops of gDBS iterations are done, and 10 and 5 pixels are treated in the first and second gDBS iterations (90 and 180 iterations). After that, two BDBS loops are performed (1800 iterations). Rapid convergence at the first stage and further optimization is achieved as in the two-mode case, although it is not shown due to the limitation of the space. Final mosaic pattern is shown at the bottom panel of Fig. 6, together with the SEM image of the fabricated device. Calculated field distributions are shown at the top panels of Fig. 6. Dashed lines in Fig. 7 shows the calculated output powers (as DMUX) for each input mode. Although the output powers are not perfectly 1/3 for each input, the modes are reasonably mixed. Solid lines in Fig. 7 shows the

measured transmission spectra of the device. The ADC for TE_2 mode (ADC2) is also added before the mosaic region to excite the mode, as shown in the micrograph of the top panel of Fig. 7. Again, good agreement with the theory can be seen, showing the proof-of-concept of the device. Further improvement can be done by changing the size of the mosaic region.

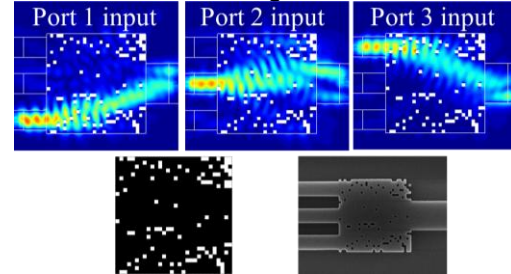


Fig 6: (Top) Calculated field distributions for each input port. (Bottom) designed mosaic pattern and SEM image of the fabricated device.

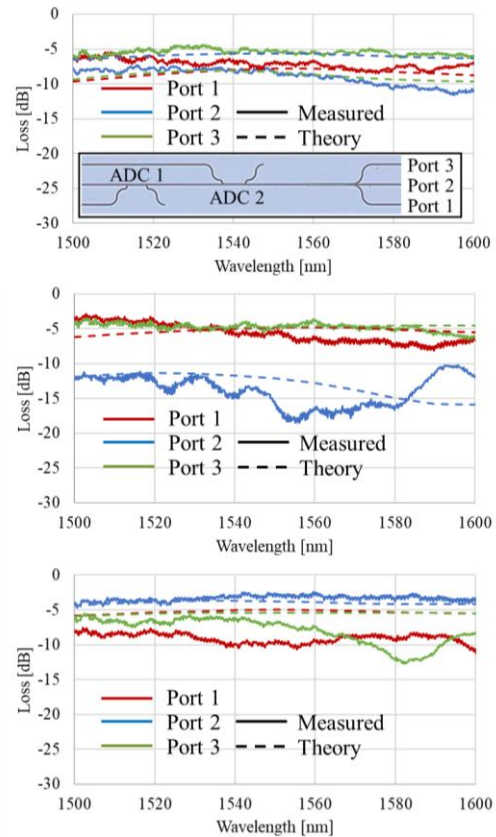


Fig 7: Measured and calculated transmission spectra of three-mode MUX for (top) TE_0 , (middle) TE_1 , and (bottom) TE_2 input.

Conclusions

The first scrambling-type mode MUXs based on mosaic structure were demonstrated. Mosaic patterns are designed with the combination of physics-oriented and machine-learning optimization techniques. Proof-of-concept devices are successfully fabricated, and they can be promising building blocks for future MDM systems.

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